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Advanced Math

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08

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

6

The correct answer is 6. It's given that a line with equation $2y = 4.5$ intersects a parabola with equation $y = -4x^2 + bx$, where b is a positive constant, at exactly one point in the xy-plane. It follows that the system of equations consisting of $2y = 4.5$ and $y = -4x^2 + bx$ has exactly one solution. Dividing both sides of the equation of the line by 2 yields $y = 2.25$. Substituting 2.25 for y in the equation of the parabola yields $2.25 = -4x^2 + bx$. Adding $4x^2$ and subtracting bx from both sides of this equation yields $4x^2 - bx + 2.25 = 0$. A quadratic equation in the form of $ax^2 + bx + c = 0$, where a, b, and c are constants, has exactly one solution when the discriminant, $b^2 - 4ac$, is equal to zero. Substituting 4 for a and 2.25 for c in the expression $b^2 - 4ac$ and setting this expression equal to 0 yields $b^2 - 4(4)(2.25) = 0$, or $b^2 - 36 = 0$. Adding 36 to each side of this equation yields $b^2 = 36$. Taking the square root of each side of this equation yields $b = \pm 6$. It's given that b is positive, so the value of b is 6.

Q2 D

D. $3x^2 + 49x + 14b$

Choice D is correct. Since each choice has a term of $3x^2$, which can be written as $3xx$, and each choice has a term of $14b$, which can be written as $72b$, the expression that has a factor of $x + 2b$, where b is a positive integer constant, can be represented as $3x + 7x + 2b$. Using the distributive property of multiplication, this expression is equivalent to $3xx + 2b + 7x + 2b$, or $3x^2 + 6xb + 7x + 14b$. Combining the x-terms in this expression yields $3x^2 + 7 + 6bx + 14b$. It follows that the coefficient of the x-term is equal to $7 + 6b$. Thus, from the given choices, $7 + 6b$ must be equal to 7, 28, 42, or 49. Therefore, $6b$ must be equal to 0, 21, 35, or 42, respectively, and b must be equal to $\frac{0}{6}$, $\frac{21}{6}$, $\frac{35}{6}$, or $\frac{42}{6}$, respectively. Of these four values of b, only $\frac{42}{6}$, or 7, is a positive integer. It follows that $7 + 6b$ must be equal to 49 because this is the only choice for which the value of b is a positive integer constant. Therefore, the expression that has a factor of $x + 2b$ is $3x^2 + 49x + 14b$.

Choice A is incorrect. If this expression has a factor of $x + 2b$, then the value of b is 0, which isn't positive.

Choice B is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{21}{6}$, which isn't an integer.

Choice C is incorrect. If this expression has a factor of $x + 2b$, then the value of b is $\frac{35}{6}$, which isn't an integer.

Q3

A

A. $y = 331{,}776^{x + 1}$

Choice A is correct. The y-intercept of a graph in the xy-plane is the point where $x = 0$. Substituting 0 for x in the given equation, $y = 576^{2x + 2}$, yields $y = 576^{20 + 2}$, which is equivalent to $y = 576^2$, or $y = 331{,}776$. Therefore, the graph of the given equation in the xy-plane has a y-intercept of $0, 331{,}776$. It follows that $r = 0$ and $s = 331{,}776$. Thus, the equivalent equation $y = 331{,}776^{x + 1}$ displays the value of s as the base.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4

31

The correct answer is 31. Subtracting 7 from both sides of the equation $x^2 + 6x + 7 = 0$ yields $x^2 + 6x = -7$. To complete the square, adding $\frac{6}{2}^2$, or 3^2, to both sides of this equation yields $x^2 + 6x + 3^2 = -7 + 3^2$, or $x + 3^2 = 2$. Taking the square root of both sides of this equation yields $x + 3 = \pm\sqrt{2}$. Subtracting 3 from both sides of this equation yields $x = -3 \pm \sqrt{2}$. Therefore, the solutions r and s to the equation $x^2 + 6x + 7 = 0$ are $-3 - \sqrt{2}$ and $-3 + \sqrt{2}$. Since $r < s$, it follows that $r = -3 - \sqrt{2}$ and $s = -3 + \sqrt{2}$. Subtracting 8 from both sides of the equation $x^2 + 8x + 8 = 0$ yields $x^2 + 8x = -8$. To complete the square, adding $\frac{8}{2}^2$, or 4^2, to both sides of this equation yields $x^2 + 8x + 4^2 = -8 + 4^2$, or $x + 4^2 = 8$. Taking the square root of both sides of this equation yields $x + 4 = \pm\sqrt{8}$, or $x + 4 = \pm 2\sqrt{2}$. Subtracting 4 from both sides of this equation yields $x = -4 \pm 2\sqrt{2}$. Therefore, the solutions t and u to the equation $x^2 + 8x + 8 = 0$ are $-4 - 2\sqrt{2}$ and $-4 + 2\sqrt{2}$. Since $t < u$, it follows that $t = -4 - 2\sqrt{2}$ and $u = -4 + 2\sqrt{2}$. It's given that the solutions to $x^2 + 14x + c = 0$, where c is a constant, are $r + t$ and $s + u$. It follows that this equation can be written as $x - r + tx - s + u = 0$, which is equivalent to $x^2 - r + t + s + ux + r + ts + u = 0$. Therefore, the value of c is $r + ts + u$. Substituting $-3 - \sqrt{2}$ for r, $-4 - 2\sqrt{2}$ for t, $-3 + \sqrt{2}$ for s, and $-4 + 2\sqrt{2}$ for u in this equation yields $-3 - \sqrt{2} + -4 - 2\sqrt{2}\ -3 + \sqrt{2} + -4 + 2\sqrt{2}$, which is equivalent to $-7 - 3\sqrt{2}\ -7 + 3\sqrt{2}$, or $-7-7 - 3\sqrt{2}\ 3\sqrt{2}$, which is equivalent to $49 - 18$, or 31. Therefore, the value of c is 31.

Q5

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Answer not provided in source data — see rationale below.

The correct answer is $\frac{7}{6}$. The value of $\frac{a}{b}$ can be found by first rewriting the left-hand side of the given equation as $\frac{x^{\frac{5}{2}}}{x^{\frac{4}{3}}}$. Using the properties of exponents, this expression can be rewritten as $x^{\left(\frac{5}{2}-\frac{4}{3}\right)}$. This expression can be rewritten by subtracting the fractions in the exponent, which yields $x^{\frac{7}{6}}$. Thus, $\frac{a}{b}$ is $\frac{7}{6}$. Note that 7/6, 1.166, and 1.167 are examples of ways to enter a correct answer.

Q6

168

The correct answer is 168. The quadratic function g gives the estimated depth of the seal, gt, in meters, t minutes after the seal enters the water. It's given that function g estimates that the seal reached its maximum depth of 302.4 meters 6 minutes after it entered the water. Therefore, function g can be expressed in vertex form as $gt = at - 6^2 + 302.4$, where a is a constant. Since it's also given that the seal reached the surface of the water after 12 minutes, $g12 = 0$. Substituting 12 for t and 0 for gt in $gt = at - 6^2 + 302.4$ yields $0 = a12 - 6^2 + 302.4$, or $36a = -302.4$. Dividing both sides of this equation by 36 gives $a = -8.4$. Substituting -8.4 for a in $gt = at - 6^2 + 302.4$ gives $gt = -8.4t - 6^2 + 302.4$. Substituting 10 for t in gt gives $g10 = -8.410 - 6^2 + 302.4$, which is equivalent to $g10 = -8.44^2 + 302.4$, or $g10 = 168$. Therefore, the estimated depth, to the nearest meter, of the seal 10 minutes after it entered the water was 168 meters.

Q7

A

A. -6

Choice A is correct. It's given that $y = x + 9$ and $y = x^2 + 16x + 63$; therefore, it follows that $x + 9 = x^2 + 16x + 63$. This equation can be rewritten as $x + 9 = x + 9x + 7$. Subtracting $x + 9$ from both sides of this equation yields $0 = x + 9x + 7 - x + 9$. This equation can be rewritten as $0 = x + 9x + 7 - 1$, or $0 = x + 9x + 6$. By the zero product property, $x + 9 = 0$ or $x + 6 = 0$. Subtracting 9 from both sides of the equation $x + 9 = 0$ yields $x = -9$. Subtracting 6 from both sides of the equation $x + 6 = 0$ yields $x = -6$. Therefore, the given system of equations has solutions, x, y, that occur when $x = -9$ and $x = -6$. Since -6 is greater than -9, the greatest possible value of x is -6.

Choice B is incorrect. This is the negative of the greatest possible value of x when $y = 0$ for the second equation in the given system of equations.

Choice C is incorrect. This is the value of y when $x = 0$ for the first equation in the given system of equations.

Choice D is incorrect. This is the value of y when $x = 0$ for the second equation in the given system of equations.

Q8

B

B. 7

Choice B is correct. Using the distributive property, the first given expression can be rewritten as $6x^2 + 3px + 24 – 16px – 64x + 24$, and then rewritten as $6x^2 + (3p – 16p – 64)x + 24$. Since the expression $6x^2 + (3p – 16p – 64)x + 24$ is equivalent to $6x^2 – 155x + 24$, the coefficients of the x terms from each expression are equivalent to each other; thus $3p – 16p – 64 = –155$. Combining like terms gives $–13p – 64 = –155$. Adding 64 to both sides of the equation gives $–13p = –71$. Dividing both sides of the equation by $–13$ yields $p = 7$.

Choice A is incorrect. If $p = –3$, then the first expression would be equivalent to $6x^2 – 25x + 24$. Choice C is incorrect. If $p = 13$, then the first expression would be equivalent to $6x^2 – 233x + 24$. Choice D is incorrect. If $p = 155$, then the first expression would be equivalent to $6x^2 – 2{,}079x + 24$.

Q9

B

B. 3

Choice B is correct. An x-intercept of a graph in the xy-plane is a point on the graph where the value of y is 0. Substituting 0 for y in the given equation yields $0 = 2x - dx + dx + gx - d$. By the zero product property, the solutions to this equation are $x = d$, $x = -d$, $x = -g$, and $x = d$. However, $x = d$ and $x = d$ are identical. It's given that d and g are unique positive constants. It follows that the equation $0 = 2x - dx + dx + gx - d$ has 3 unique solutions: $x = d$, $x = -d$, and $x = -g$. Thus, the graph of the given equation has 3 distinct x-intercepts.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q10

-10

The correct answer is -10. Adding y to both sides of the second equation in the given system yields $20x + 100 = y$. Substituting $20x + 100$ for y in the first equation in the given system yields $x^2 + 20x + 100 + 7 = 7$. Subtracting 7 from both sides of this equation yields $x^2 + 20x + 100 = 0$. Factoring the left-hand side of this equation yields $x + 10x + 10 = 0$, or $x + 10^2 = 0$. Taking the square root of both sides of this equation yields $x + 10 = 0$. Subtracting 10 from both sides of this equation yields $x = -10$. Therefore, the value of x is -10.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

Answers recovered from rationale for Q5 — the source CB data omitted a structured key for these items, so the answer slot shows –. The worked solution in the rationale below each entry contains the correct value.

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